Joseph Walton

AI Engineer | Generative AI | LLMs | Enterprise AI Solutions

Email: jwgwalton@gmail.com
Telephone: 07817685781
LinkedIn: linkedin.com/in/joseph-walton-2a508591
GitHub: github.com/jwgwalton

Professional Summary

AI and machine learning engineer with more than 10 years of software engineering experience, delivering production AI solutions across central government, intellectual property, financial services and defence. Specialist in generative AI, LLM agents, RAG, semantic and vector search, evaluation and observability. Proven record of building secure enterprise applications on Azure, integrating AI services within service-oriented architectures and taking solutions from discovery and design through deployment, monitoring and continuous improvement. Trusted technical leader who collaborates effectively with business stakeholders, architects and cross-functional engineering teams, with a strong focus on responsible AI and measurable business outcomes.

Core Skills

- **Generative AI & LLMs:** Azure OpenAI, Azure AI Foundry, LangChain, LangGraph, Claude Skills, Hugging Face, RAG, AI agents, prompt evaluation
- **Search & Retrieval:** Semantic search, vector search, Elasticsearch, HNSW, recommendation systems
- **Programming & APIs:** Python, SQL, Java, Azure Functions, service-oriented architectures
- **Cloud & Engineering:** Microsoft Azure, Google Cloud Platform, AWS, Docker, Git, CI/CD, MLOps, MLflow
- **Data & Machine Learning:** Polars, Pandas, PySpark, TensorFlow, PyTorch, Scikit-learn, NLP
- **Responsible AI:** Model evaluation, observability, red teaming, AI governance, human-in-the-loop workflows

---

Professional Experience

IBM

Lead Data Scientist — Central Government Contract
June 2025 - Present

- Rescued underperforming enterprise AI projects by introducing robust evaluation and observability practices, improving delivery confidence and production readiness.
- Designed and delivered an LLM-powered data processing agent using Azure AI Foundry, Azure Functions and LangGraph, saving case workers two hours per day and approximately £2M annually.
- Created a Claude Skill to streamline human-in-the-loop ingestion of evaluation data for LLM products, saving approximately one hour for each case added to the evaluation suite.
- Re-engineered a data matching workflow from Apache Spark to Polars, reducing runtime from two hours to ten minutes and cloud costs by approximately £10k per month.
- Worked across technical and business teams to turn operational requirements into maintainable, measurable AI solutions for a secure government environment.

Tech Stack: Python, Azure AI Foundry, Azure Functions, LangGraph, Claude, Polars, Apache Spark

---

Intellectual Property Office

Senior Machine Learning Engineer

November 2021 - April 2025

- Set the product roadmap for AI applications in the intellectual property domain, balancing business value, technical feasibility, security and responsible AI considerations.
- Designed and developed generative AI applications within a service-oriented architecture using Azure OpenAI, LangChain and Azure Functions.
- Built a RAG agent over specialist enterprise knowledge to help novice users identify relevant Goods and Services terms.
- Implemented Elasticsearch HNSW vector search and systematically tuned throughput and recall for production workloads.
- Improved search and recommendation relevance by 20% through metric-led evaluation and iterative refinement.
- Conducted red-team exercises to identify adversarial risks and improve the robustness of generative AI applications.
- Implemented an end-to-end MLOps strategy across multiple environments, reducing model deployment time to under one day.
- Architected resilient Polars ETL services to replace Azure Databricks workloads, cutting compute costs by 50% and enabling zero-downtime deployments.
- Presented the organisation's AI approach to technical and non-technical stakeholders at international IP offices and contributed to cross-government AI policy and best practices.

Tech Stack: Python, Azure OpenAI, LangChain, Hugging Face, Azure Functions, Azure ML Pipelines, Elasticsearch, HNSW, Polars

---

Incopro

Lead Machine Learning Engineer

February 2020 - November 2021

- Led cross-functional delivery of production machine learning solutions using large-scale multilingual text and image data.
- Designed an MLOps framework on Google Cloud Platform that orchestrated training, evaluation, approval and deployment with Vertex AI Pipelines, reducing time to market to under one day.
- Developed and deployed a multilingual BERT model for detecting intellectual property infringements across more than 90 languages.
- Built a recommendation engine combining text and image similarity to improve IP violation detection.
- Mentored a software engineer through a transition into machine learning engineering.

Machine Learning Engineer

November 2018 - February 2020

- Built and deployed a secure, containerised NLP system for predicting IP infringement, increasing enforced infringements by 10% per analyst across approximately 200 analysts.
- Delivered an image similarity model that helped secure a blue-chip client contract representing 10% of annual recurring revenue.

Tech Stack: Python, Hugging Face, TensorFlow, PySpark, Vertex AI Pipelines, Google Cloud Platform, MLflow, Docker

---

Office for National Statistics

Machine Learning Engineer
September 2017 - November 2018

- Developed machine learning models and scalable data pipelines to link and process government administrative data securely in support of national surveys.
- Worked in a cross-functional team on the UK Census 2021 programme, developing a highly scalable backend for the census data collection system.

Tech Stack: Python, PySpark, Cloudera

Earlier Experience

Barclays — Data Scientist (Secondment) | *September 2018 - October 2018*
Developed regional economic indicators from payments data as a continuation of the winning ONS x Barclays Hackathon project.

Purple Secure Systems — Software Engineer | *August 2014 - October 2016*
Developed full-stack big data processing systems in Java, Python, PySpark and AngularJS for government and defence clients.

Education

University of Bath

Master of Science in Applied Mathematics | *2016 - 2017*
Bachelor of Science in Physics | *2010 - 2013*

Award

Winner, ONS x Barclays Data Hackathon — Used payments data to create more granular and timely estimates for Gross Value Added (GVA).